Transmission Control Protocol (TCP) Architecture

Coverage: TCP 20-Byte Header Format, 3-Way Handshake Connection Setup, 4-Way Connection Teardown (TIME_WAIT 2MSL), Sequence/ACK Mechanics

Prepared by: Prof. Anju Markose | Assistant Professor, Dept. of Computer Science & Engineering | VJCET

1. TCP Header & Flags

- 20-60 Byte Header: Seq Num (32b), Ack Num (32b), Flags (SYN, ACK, FIN, RST, PSH, URG), Window Size (16b), Checksum (16b).
- Cumulative ACK: Confirms receipt of all bytes up to Ack - 1.

2. Connection Lifecycle

- 3-Way Handshake: SYN (Seq=x) -> SYN-ACK (Seq=y, Ack=x+1) -> ACK (Seq=x+1, Ack=y+1).
- 4-Way Teardown: FIN -> ACK -> FIN -> ACK. TIME_WAIT (2MSL) ensures final ACK delivery.

■ KTU University High-Yield Exam Questions

- Q: Draw the TCP header and explain the 3-Way Handshake with Sequence and Ack numbers. (8 Marks)